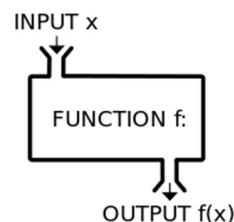
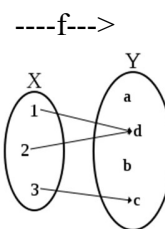


- (a) A function (函數) associates one quantity, also known as the input (輸入), with another quantity called the output (輸出). The processed outcome of an input x is called the image (像) of x , and may be written as $f(x)$, or denoted as y .

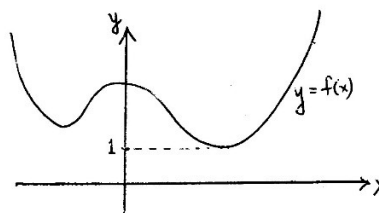


- (b) The set of all inputs to a function is called the domain (定義域) of the function. A set which contains all possible outputs is called the co-domain (值域). The set of all outputs of a function is called its image set (像集).

For example: Here domain $f = \{1, 2, 3\}$
 Co-domain $f = \{a, b, c, d\}$
 $f(1) = d, f(2) = d, f(3) = c$
 Image $f = \{c, d\}$



- (c) If both the domain and co-domain of the function are the set of all real numbers, then we can represent the function in the form of a graph. Here Image $f = \{x \in \mathbb{R} : x \geq 1\}$



Functions with Special Properties

Def.: 1. $f(x)$ is called an odd function (奇函數) if $f(-x) = -f(x)$ for all x in the domain.

For example: $f(x) = x^3, f(x) = \sin(x)$

2. $f(x)$ is called even function (偶函數) if $f(-x) = f(x)$ for all x in the domain.

For example: $f(x) = x^2, f(x) = \cos(x)$

3. $f(x)$ is called a periodic function (週期函數) if there exists a number k such that for all $x, f(x) = f(x + k)$. For example: $f(x) = \tan(x)$.

Example 1: Given that $f(x)$ is a function from $\mathbb{R}$ to $\mathbb{R}$ and that for some $T \in \mathbb{R}, f(T) = 1$ and $f(x) > 0$ for all $x \in (0, 2T)$. Further, $f(a-b) = \frac{f(a)-f(b)}{1+f(a)f(b)}$ for any $a, b \in \mathbb{R}$,

- Show that $f(x)$ is an odd function.
- Show that $f(x)$ is monotonic increasing in $(0, 2T)$.
- Show that $f(x)$ is a periodic function and find its period.

Example 2: Given that $f(x)$ is a function from $\mathbb{R}$ to $\mathbb{R}$ and that $f(a) = 0$ for some $a \in \mathbb{R}$. Further, $f(x+y) + f(x-y) = 2f(x)f(y)$ for any $x, y \in \mathbb{R}$,

- Show that $f(x)$ is an even function.
- Show that $f(x)$ is a periodic function and find its period.
- Find $f(na)$ for any integer n .

Example 3: Given that $f(x)$ is a function from $\mathbb{R}$ to $\mathbb{R}$. If a non-zero real numbers 'a' exists such that for any value of x , the following equations are satisfied. In each individual case, show that $f(x)$ is a periodic function, and find the period.

- (i) $f(x) + f(x+a) = 0$,
- (ii) $f(x) = f(x-a)f(x+a)$,
- (iii) $f(x+a) = \frac{1-f(x)}{1+f(x)}$,
- (iv) $f(x+a) = \frac{1+f(x)}{1-f(x)}$,
- (v) $f(x+a) = \frac{1}{f(x)}$,
- (vi) $f(x+a) = -\frac{1}{f(x)}$.

Example 4: Let f be a real valued function defined for all real numbers x such that, for some positive constant a , the equation $f(x+a) = \frac{1}{2} + \sqrt{f(x) - [f(x)]^2}$ holds for all x .

- (a) Prove that the function f is periodic (i.e., there exists a positive number b such that $f(x+b) = f(x)$ for all x).
- (b) For $a = 1$, give an example of a non-constant function with the required properties. (10th IMO 1968 No. 5)

Functional Equations (函數方程)

An equation which contains one or more unknown functions is called a functional equation.

Methods of Solving Functional equations

(a) Change of Variable (換元法)

Method: Change the given variable to a new one to obtain new equation(s), then group the new equation(s) and the original equation to obtain a new system of functional equations and then solve the system of equations.

Example 5: If f is a function from $\mathbb{R}$ to $\mathbb{R}$ and $x^2f(x) + f(1-x) = 2x - x^4$, find $f(x)$.

Example 6: If $\left(\frac{1}{x}\right)f(-x) + f\left(\frac{1}{x}\right) = x$ for all $x \in \mathbb{R}$, find $f(x)$.

Example 7: Solve $f(x) + f\left(1 - \frac{1}{x}\right) = 1 + x$ for $x \neq 0, 1$.

Note: In general, for functions equations in the form $a(x)f(x) + b(x)f[g(x)] = c(x)$, where $g(x)$ is a function such that $g^{(k)}(x) = x$, then try replacing x by $g(x)$.

Example 8: If f is a function from $\mathbb{R}$ to $\mathbb{R}$ and $f(x) + xf(1-x) = 1 + x$, find $f(x)$. Will the answer be different if f is a function defined on the set of complex numbers?

(b) Finding value of function at some particular points (取特殊值法)

Method: Check the format and trend of the function when the independent variable takes some particular values.

Example 9: Given $f(x)$ is a function defined on the set of real numbers such that

$$f(x) = 1 - \frac{1}{2}[f(x-1) - f(x-2)], \text{ and } f(1) = \frac{1}{2}, f(2) = \frac{3}{4}, \text{ find } f(x).$$

Example 10: Given $f(x)$ is a function defined on the set of real numbers such that

$$f(x)f(y) - f(xy) = x+y \text{ for any } x, y \in \mathbb{R}. \text{ Find } f(x).$$

Example 11: Given $f(x)$ is a function defined on the set of real numbers such that

$$xf(y) + yf(x) = (x+y)f(x)f(y) \text{ for any } x, y \in \mathbb{R}. \text{ Find } f(x).$$

Example 12: Given $f(x)$ is a function defined on the set of real numbers such that

$$f(x+y^2) = f(x) + 2(f(y))^2 \text{ for any } x, y \in \mathbb{R}, \text{ and } f(1) \neq 0, \text{ Find } f(2019).$$

Example 13: Given $f(x)$ is a function defined on the set of positive real numbers such that

$$f(af(b)) = ab \text{ and } f(a) \geq 0 \text{ for any } a, b \in \mathbb{R}^+. \text{ Find } f(2019).$$

Example 14: Given $f(x)$ is a function defined on the set of real numbers such that

$$f(x) + (x + \frac{1}{2})f(1-x) = 1 \text{ for any } x \in \mathbb{R}. \text{ Find } f(0), f(1) \text{ and } f(x).$$

Example 15: The function $f(x, y)$ satisfies (i) $f(0, y) = y+1$

$$(ii) \quad f(x+1, 0) = f(x, 1)$$

$$(iii) \quad f(x+1, y+1) = f(x, f(x+1, y))$$

for all non-negative integers x, y . Determine $f(4, 1981)$

(22nd IMO 1981, No. 6)

(c) Finding a constant quantity (找尋常數)

Method: Check the format of the function when the dependent variable takes some fixed values.

Example 16: Given that $f(x)$ is a function from $\mathbb{R}$ to $\mathbb{R}$, and $f[f(x) + y] = f(x+y) + f(0)$ for all $x \in \mathbb{R}$, find $f(x)$.

Example 17: Given $f(x)$ is a function defined on the set of real numbers such that

$$(x-y)f(x+y) - (x+y)f(x-y) = 4xy(x^2 - y^2) \text{ for any } x, y \in \mathbb{R}. \text{ Find } f(x).$$

Example 18: Given $f(x)$ is a function defined on the set of real numbers such that

$$f(xf(y)) = f(xy) + x \text{ for any } x, y \in \mathbb{R}. \text{ Find } f(x).$$

(d) Cauchy's Functional Equation (柯西函數方程)

Method: Observe the changes when the independent variable takes the value of some particular natural numbers. Then extend the investigation to integers, rational numbers, and finally real numbers.

Example 19: Given $f(x)$ is a function over the set of rational numbers such that for all rational numbers x, y , $f(x + y) = f(x) + f(y)$ (Cauchy Equation)

Example 20: Solve the following equation $f(x + y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$

Example 21: Solve the equation $f\left(\frac{x+y}{2}\right) = \frac{f(x)+f(y)}{2}$ for all $x, y \in \mathbb{R}$

Example 22: Solve the equation $f(xy) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$

Example 23: Solve the equation $f(xy) = f(x)f(y)$ for all $x, y \in \mathbb{R}$

More Problems

Example 24: Find all polynomials P , in two variables, with the following properties:

- (i) for a positive integer n and all real t, x, y , $P(tx, ty) = t^n P(x, y)$
- (ii) for all real a, b, c , $P(b+c, a) + P(c+a, b) + P(a+b, c) = 0$
- (iii) $P(1, 0) = 1$

(17th IMO 1975, No. 6)

Example 25: Find all functions f , defined on the non-negative real numbers and taking non-negative real values, such that

- (i) $f(x+y) = f(x)f(y) + f(y)$, for all $x, y \geq 0$
- (ii) $f(2) = 0$ and
- (iii) $f(x) \neq 0$ for $0 \leq x < 2$.

(27th IMO 1986, No.5)

Example 26: Let $\mathbb{Q}^+$ be the set of positive rational numbers. Construct a function

$$f: \mathbb{Q}^+ \rightarrow \mathbb{Q}^+ \text{ such that } f(xf(y)) = \frac{f(x)}{y} \text{ for all } x, y \text{ in } \mathbb{Q}^+.$$

(31st IMO 1990, No. 4)

Example 27: Let S be the set of real numbers strictly greater than -1 . Find all functions $f: S \rightarrow S$ satisfying the two conditions:

1. $f(x + f(y) + xf(y)) = y + f(x) + xf(x)$ for all x and y in S ;
2. $\frac{f(x)}{x}$ is strictly increasing on each of the intervals $-1 < x < 0$ and $0 < x$.

(35th IMO 1994, No. 5)

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